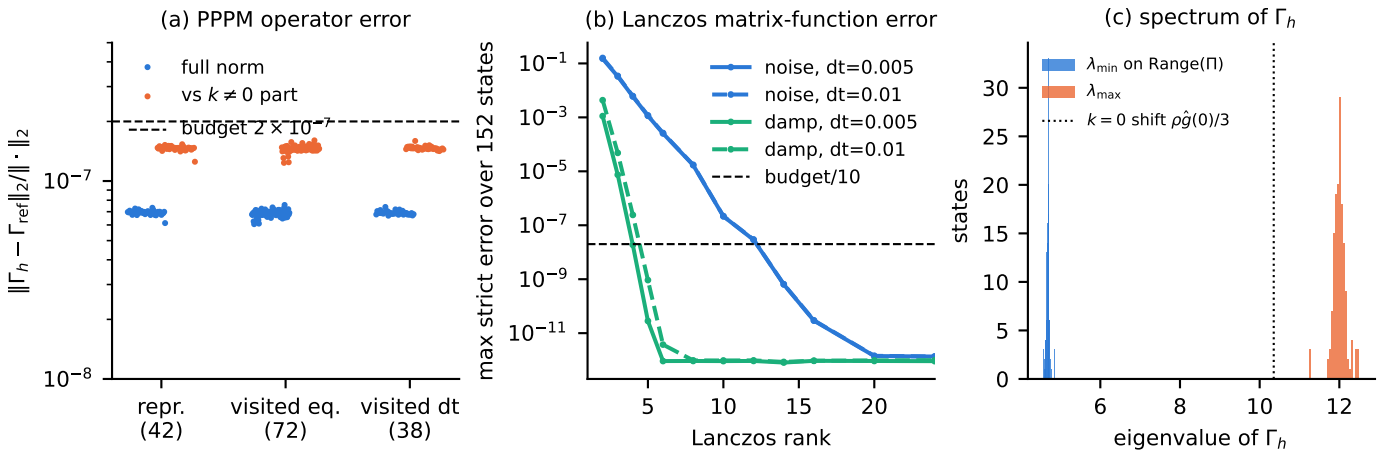
(a) PPPM operator error
full norm
vs $k \neq 0$ part
budget 2×10^{-7}
$\|\Gamma_h - \Gamma_{\mathrm{ref}}\|_2 / \|\cdot\|_2$
10^{-7}
10^{-8}
repr. (42)
visited eq. (72)
visited dt (38)
(b) Lanczos matrix-function error
noise, dt=0.005
noise, dt=0.01
damp, dt=0.005
damp, dt=0.01
budget/10
max strict error over 152 states
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
5
10
15
20
Lanczos rank
(c) spectrum of Γ_h
$\lambda_{\min}$ on Range(Π)
$\lambda_{\max}$
$k = 0$ shift $\rho\hat{g}(0)/3$
states
0
5
10
15
20
25
30
6
8
10
12
eigenvalue of Γ_h